Physical signals and imaging modalities

Imaging

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Questions for each modality:

- **Physical signal:** waves or particles that carry information from the body to the detector.
  - What signal does this modality use?
- **Detector measurement:** the quantity recorded from the received signal.
  - What does the detector record?
- **Contrast mechanism:** tissue properties or processes that make image regions differ.
  - Why do different tissues or regions give different image values?

Object, measurements, and image

An **image** is a map of measured or estimated values arranged by spatial position.

Stage	Example in X-ray Computed Tomography (CT)
Object property	How strongly tissue attenuates X-rays
Measurement	Transmitted X-ray signal along a path
Image	Estimated attenuation at locations in a slice

The measurement and the reconstructed image describe different quantities. Image reconstruction may be simple or mathematically complex.

Imaging chain

Stage	Visible-light camera example
Source	Light illuminating the scene
Interaction	Objects reflect or emit light
Propagation and collection	Light travels to the camera and passes through its lens
Detection	Sensor elements convert light into electrical signals

The source, the object, and the detector each affect the recorded signal.

Imaging chain part 2

Stage	Visible-light camera example
Processing or reconstruction	Convert detector data into image values
Display	Map image values to brightness and color
Perception and interpretation	A viewer identifies structures or differences

Data transfer and storage can occur between stages. Reconstruction may require many measurements, as in CT.

Radiography and CT

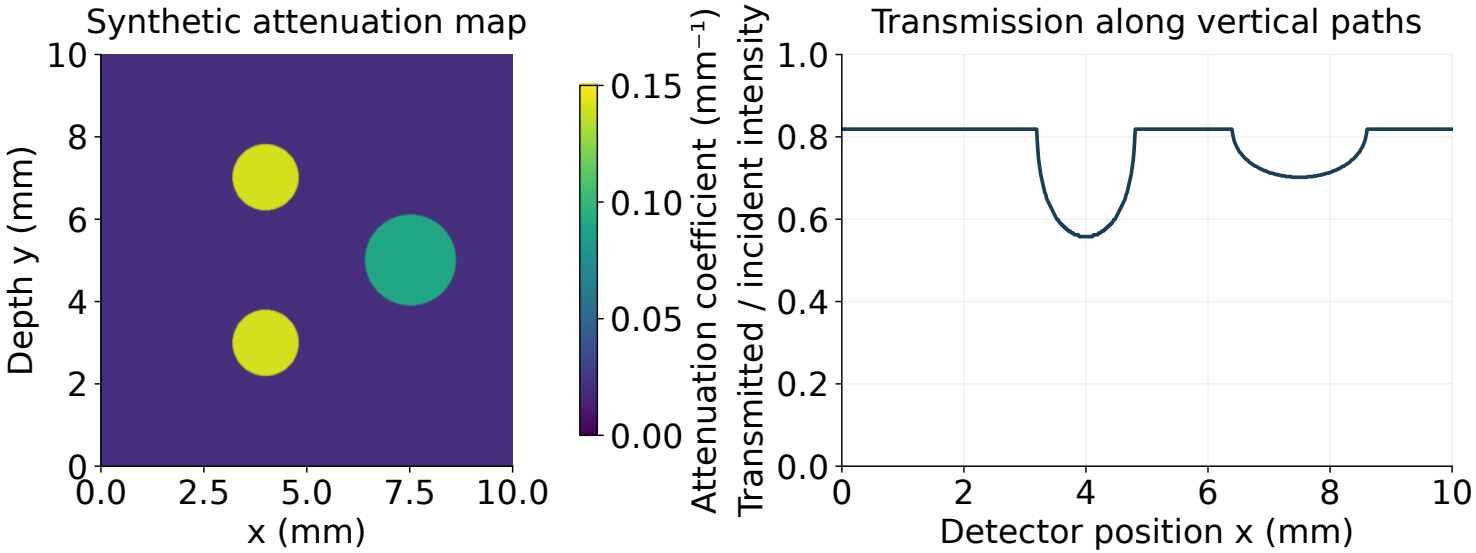
Physical signal: X-ray photons transmitted through the body.

Method	Detector measurement	Image formation
Radiography	Transmitted X-ray intensity across a detector	A projection with overlapping structures
CT	Transmitted X-ray intensity from many directions	Reconstruction of cross-sectional slices

Contrast mechanism: differences in X-ray attenuation, the reduction of the primary beam as it passes through material.

Projection imaging

A **projection** records the combined effect of material along each measurement path. In radiography, the transmitted X-ray intensity depends on attenuation along the entire path through the body.



Can this measurement tell us where along the path the attenuation occurred?

Ultrasound

A transducer sends a sound pulse (acoustic energy) into tissue and detects returning echoes.

- **Physical signal:** reflected sound waves, or echoes.
- **Detector measurement:** echo intensity (via voltage) as a function of time.
- **Contrast mechanism:** differences in sound reflection and propagation between tissue regions.

Echo delay helps locate a reflecting boundary. The transducer converts the returning sound into an electrical signal.

Magnetic resonance imaging (MRI)

MRI measures the response of nuclei in tissue to radiofrequency excitation in a magnetic field.

- **Physical signal:** radiofrequency electromagnetic signal from nuclei in tissue.
- **Detector measurement:** voltage in a receiver coil.
- **Contrast mechanism:** differences in the strength and decay of the tissue's magnetic response.

The scanner uses spatial variations (gradients) in an applied magnetic field to determine where the signal originates.

Positron emission tomography (PET)

A **radioactive tracer** is introduced into the body to follow a biological process.

- **Physical signal:** photon pair produced when a positron from the tracer annihilates with an electron.
- **Detector measurement:** pairs of photon detections close together in time.
- **Contrast mechanism:** differences in the distribution of the tracer.

Many detection pairs allow the scanner to estimate where the tracer is located.

Physical frequency and spatial frequency

Physical wave frequency: oscillations per second, in hertz (Hz).

Examples include an ultrasound pressure wave and an MRI radiofrequency signal.

Spatial frequency: repetitions per unit distance, such as cycles/mm.

Example: closely spaced bright and dark stripes in an image have a high spatial frequency.

These frequencies describe different variables, time and position.

Wavelength, frequency, and photon energy

For electromagnetic radiation in free space,

$$c = \lambda\nu, \qquad E_\gamma = h\nu = \frac{hc}{\lambda}.$$

- λ: wavelength in m; ν: temporal frequency in Hz.
- E_γ: energy of **one photon**, in J.
- $c \approx 3.00 \times 10^8$ m/s; $h \approx 6.626 \times 10^{-34}$ J s.
  - For a photon, Joules are extremely small. Instead the eV.

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}, \qquad E_\gamma[\text{eV}] \approx \frac{1240}{\lambda[\text{nm}]}.$$

Photon calculations and stated assumptions

Assumed signal	Free-space wavelength	Energy per photon
Visible light at 500 nm	500 nm	2.48 eV
RF at 100 MHz	3.00 m	4.14×10^{-7} eV
X-ray photon at 50 keV	0.0248 nm	50 keV

$$1 \text{ eV} = 1.602 \times 10^{-19} \text{ J}, \qquad E_\gamma[\text{eV}] \approx \frac{1240}{\lambda[\text{nm}]}.$$

Note: An X-ray beam contains a spectrum of photon energies centered around an average energy.

Pixel values

A **pixel** is one element of a digital image. Its value represents a measurement or an estimated quantity over a spatial region.

- A reconstructed image can have different units from the detector signal.
- Display settings map numerical values to gray levels or colors.

Repeated photon counts: Noise

Repeated exposures under the same conditions can give different detected counts.

For independent arrivals with a constant expected count $\overline{N}$, a model with noise from Poisson statics is

$$N \sim \text{Poisson}(\overline{N}), \qquad \text{E}[N] = \overline{N}, \qquad \text{Var}(N) = \overline{N}.$$

The standard deviation is $\sigma_N = \sqrt{\overline{N}}$.

An observed count N is one outcome. The expected count $\overline{N}$ describes the mean over repeated exposures.

SNR for Poisson counting

Here, **signal-to-noise ratio (SNR)** means the expected count divided by its standard deviation:

$$\text{SNR} = \frac{\overline{N}}{\sigma_N} = \sqrt{\overline{N}}.$$

For $\overline{N} = 100$, $\sigma_N = 10$ counts and SNR $= 10$.

Predict: if the expected count increases by a factor of four, what happens to the standard deviation and SNR?

Pixel spacing and field of view

The **field of view (FOV)** is the physical extent represented by the image.

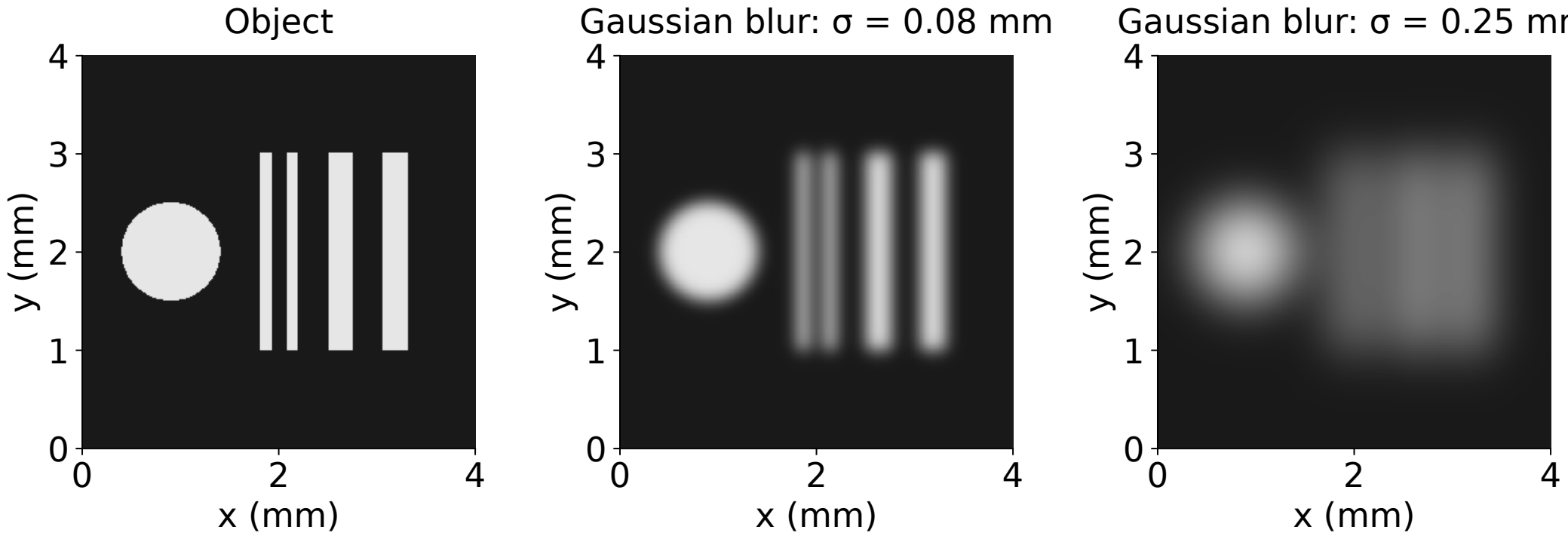
The **pixel spacing**, also called pitch, is the distance between neighboring pixel centers.

For N equally spaced pixels spanning width L,

$$p = \frac{L}{N}.$$

Example: $L = 128$ mm and $N = 256$ give $p = 0.50$ mm.

Spatial detail and blur



Each panel images the same physical object at the same intensity.

Contrast and resolution

Contrast describes differences in recorded or displayed values between regions.

Spatial resolution describes the ability to distinguish nearby structures or fine spatial detail.

A feature can become difficult to see because blur reduces its contrast. Small pixel spacing alone does not guarantee fine resolution.

In-class discussion

For each example, let's identify the physical signal and detector measurement.

1. An X-ray detector records a weaker transmitted signal at one location.
2. An ultrasound probe receives an echo later than a previous echo. What causes echoes?
3. A PET scanner records a pair of photon detections on opposite sides of the scanner.

Which measurements directly identify a depth? Which need additional measurements or assumptions to fully reconstruct the object?

Image formation in each modality

Modality	Physical signal	How position is determined
Visible-light camera	Reflected or emitted visible light	Lens maps light directions to sensor positions
Radiography	Transmitted X-ray photons	Projection geometry
CT	Transmitted X-ray photons	Many projection directions
Ultrasound	Reflected sound waves	Echo timing and beam direction
MRI	Radiofrequency electromagnetic signal	Magnetic-field gradients
PET	Annihilation photons	Coincidence measurements

Questions to consider before next time

1. What is the difference between detector data and a reconstructed image?
2. At a fixed field of view, what happens to pixel spacing when the pixel count doubles in each direction?
3. Why can an image with many pixels still appear blurred?

Next meeting: spatial signals, Fourier representations, and sampling.

Reading and resources

Prince and Links, *Medical Imaging: Signals and Systems*: introductory material on imaging modalities and image representation.

- NIBIB: Medical X-rays
- NIBIB: Computed tomography
- NIBIB: Ultrasound
- NIBIB: MRI
- NIBIB: Nuclear medicine